Applied Machine Learning Group Project

The group project is your opportunity to demonstrate what you've learned in the course and apply it to a new scenario. The project should address a novel problem, offering a unique solution or exploring an under-explored scenario. The problem you choose should be sufficiently complex—avoid simple classification tasks using well-known datasets.

The group is responsible for sourcing an appropriate dataset. This requires careful consideration of data size, relevance to the problem, and feasibility. Assess the dataset's quality early on to ensure it is clean (or clean enough) and has the necessary labels. If data is missing or requires preprocessing, account for that in your proposal. By the time you submit your proposal, you should have chosen and worked with a dataset; therefore, you should not change your dataset once your proposal is submitted.

Proposal Format

The project proposal is a road map on how you will execute your project. Use the following outline for your project proposal submission. Your proposal should be no longer than 2-3 pages, not including the title page, and submitted as a pdf.

Title

Choose a concise, descriptive title for your project, giving an idea of its focus. List all team members on the title page.

Introduction

This section should introduce your project with a high-level objective. Describe the business or scientific problem you are aiming to solve, including the context and current approaches to solving this problem (some research will be necessary). Explain how your contribution is novel.

Also use this section to establish *stakeholder needs*. Identify each stakeholder and describe their role in relation to your project. For stakeholders with specific but potentially non-obvious needs, clarify why they are stakeholders and explain those needs in detail (e.g., a city traffic planner may need information on temporal variation in pedestrian traffic and the density of office buildings to determine optimal traffic signal placement).

Dataset Description

Provide a link to your data, along with summary information (e.g., the number of columns, rows, and types of features). Discuss the data's provenance—how can you verify its reliability? Does it include metadata, and if not, what other information about its origins can you provide? If you anticipate needing multiple datasets, list each one and confirm their availability.

Methods

Outline the approach you plan to take, including the supervised and/or unsupervised techniques you intend to employ. Also, discuss how you will evaluate your results. Evaluation goes beyond simple metrics; it involves assessing whether the outcomes meet stakeholder needs and align with the intended application or business objectives. Justify your choices based on your dataset(s), your problem, and what you've learned in class. Include any necessary data transformations or preprocessing steps in this section.

Risks

Describe potential risks to the project, including possible pitfalls and roadblocks, along with your mitigation strategies. What will you do if some aspect of your plan doesn't work? Include any concerns or questions you would like addressed in the feedback on your proposal.

Grading

- Does your proposal include all of the sections mentioned above?
- Is the proposal well-written and easy to follow?
- Have you clearly identified your stakeholders and their needs?
- Is your proposal plan feasible, with sufficient detail to inspire confidence in your project’s success?